

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Sameer Ahmed G	Reg. No.:	192421154
Section / Batch:	B Tech IT	Date:	23-07-2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze the recurrence $T(n) = T(n/2) + \log n$ using the substitution method. Clearly define the recurrence and apply step-by-step expansion. Justify each step mathematically and derive the final complexity. Use tables or intermediate steps if required for clarity. Present the analysis clearly and interpret the efficiency of the algorithm.	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Substitution Method for Logarithmic Recurrence

Analyze the recurrence $T(n) = T(n/2) + \log n$ using the substitution method. Clearly define the recurrence and apply step-by-step expansion. Justify each step mathematically and derive the final complexity. Use tables or intermediate steps if required for clarity. Present the analysis clearly and interpret the efficiency of the algorithm.

Code of Conduct

I Sameer Ahmed G (192421154) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sameer Ahmed G

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						90

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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Substitution Method for Logarithmic Recurrence Analysis

CSA0620 – Design and Analysis of Algorithm

Guided By:
Dr. A Sairam

Presented By:-
Sameer Ahmed G - 192421154



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PROBLEM STATEMENT

- Many divide-and-conquer algorithms are represented using recurrence relations that describe the execution time recursively.
- The given recurrence, $T(n) = T(n/2) + \log n$, reduces the input size by half while performing logarithmic work at each level.
- The objective is to solve the recurrence using the substitution method by expanding it step by step.
- Each expansion should be mathematically justified until the recurrence reaches its base condition.
- The final goal is to derive the asymptotic time complexity and interpret the efficiency of the algorithm.



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ALGORITHM OVERVIEW

- The substitution method solves a recurrence by repeatedly replacing the recursive term with its expanded form.
- Every recursive call reduces the problem size from **n** to **n/2**, creating a sequence of smaller subproblems.
- At each recursive level, an additional computational cost of **log n** is incurred.
- The recursive expansion continues until the input size becomes one, which serves as the base case.
- The accumulated cost of all recursive levels is simplified using summation formulas to determine the overall complexity.



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ALGORITHM / PSEUDOCODE

Function Solve(n)

```
if  $n == 1$ 
    return  $c$ 
else
    return Solve( $n/2$ ) +  $\log(n)$ 
```

- Base case occurs when $n = 1$.
- Each recursive call halves the input size.
- Additional work per call is $\log n$.
- Expansion continues until the recursion terminates.
- Complexity is obtained by summing all logarithmic costs.



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WORKING EXAMPLE (DRY RUN)

For $n = 16$:

$$\begin{aligned}T(16) &= T(8) + 4 \\T(8) &= T(4) + 3 \\T(4) &= T(2) + 2 \\T(2) &= T(1) + 1\end{aligned}$$

Therefore,

$$\begin{aligned}T(16) &= T(1) + 4 + 3 + 2 + 1 \\&= c + 10\end{aligned}$$

General Expansion:

Step	Expression
1	$(T(n) = T(n/2) + \log n)$
2	$(= T(n/4) + \log(n/2) + \log n)$
3	$(= T(n/8) + \log(n/4) + \log(n/2) + \log n)$
k	$(= T(n/2^k) + \sum_{i=0}^{k-1} \log(n/2^i))$



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Complexity Analysis

Recursion stops when $n/2^k = 1$.

Therefore, $k = \log n$.

Sum becomes:

$$\sum_{i=0}^{\log n - 1} (\log n - i)$$

This simplifies to:

$$\frac{\log n (\log n + 1)}{2}$$

Final Time Complexity:

$$T(n) = \Theta((\log n)^2)$$



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COMPARATIVE ANALYSIS

METHOD	SUITABLE FOR	RESULT
Substitution Method	Pattern identification	$\Theta((\log n)^2)$
Iteration Method	Stepwise expansion	$\Theta((\log n)^2)$
Recursion Tree	Visualization	$\Theta((\log n)^2)$
Master Theorem	Standard forms	Not directly applicable in basic form

Key Points

- Substitution clearly reveals recursive behavior.
- Easy to verify each mathematical step.
- Provides intuition behind recurrence solving.
- Produces the same complexity as iteration.



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APPLICATIONS

- Performance analysis of divide-and-conquer algorithms that repeatedly divide the input into smaller subproblems.
- Recursive searching and hierarchical processing algorithms where input size decreases exponentially.
- Complexity analysis of binary recursive procedures used in computer science and software engineering.
- Evaluation of recursive mathematical algorithms that involve logarithmic computations at each stage.
- Used in algorithm design courses to understand recurrence solving techniques and asymptotic analysis.



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RESULTS & DISCUSSION

- Recursive expansion shows that the number of recursive calls increases only logarithmically with the input size.
- Each recursive level contributes a logarithmic computational cost, resulting in a predictable summation pattern.
- Mathematical simplification confirms that the total execution time grows proportionally to $(\log n)^2$.
- Compared to linear or quadratic algorithms, this recurrence demonstrates significantly better scalability.
- The substitution method successfully verifies both the correctness of the recurrence solution and the efficiency of the algorithm.

CONCLUSION

- The substitution method systematically expands the recurrence.
- Mathematical simplification reveals the summation pattern.
- Final time complexity is $\Theta((\log n)^2)$.
- The method provides clear proof of algorithm efficiency.
- Useful for analyzing divide-and-conquer recurrences.

REFERENCES

- Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, *Introduction to Algorithms*, 4th Edition, MIT Press.
- Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, *Fundamentals of Computer Algorithms*.
- Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, *The Design and Analysis of Computer Algorithms*.
- Donald E. Knuth, *The Art of Computer Programming, Volume 1*.

THANK YOU